

Filter Summary Report: CG,TIA,Automated,simple,Z1,ZL

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Contents

1 Examined $H(z)$ for CG TIA Automated simple Z1 ZL: $\frac{Z_1 Z_L g_m}{Z_1 g_m + 1}$

$$H(z) = \frac{Z_1 Z_L g_m}{Z_1 g_m + 1}$$

2 AP

3 BP

3.1 BP-1 $Z(s) = \left(R_1, \infty, \infty, \infty, \infty, \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L} \right)$

$$H(s) = \frac{L_L R_1 R_L g_m s}{R_1 R_L g_m + R_L + s^2 (C_L L_L R_1 R_L g_m + C_L L_L R_L) + s (L_L R_1 g_m + L_L)}$$

Parameters:

Q: $\frac{\sqrt{C_L} R_L}{\sqrt{L_L}}$
 wo: $\frac{1}{\sqrt{C_L} \sqrt{L_L}}$
 bandwidth: $\frac{1}{C_L R_L}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_1 R_L g_m}{R_1 g_m + 1}$
 Qz: None
 Wz: None

3.2 BP-2 $Z(s) = \left(L_1 s, \infty, \infty, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{L_1 R_L g_m s}{C_L L_1 R_L g_m s^2 + s (C_L R_L + L_1 g_m) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_L} \sqrt{L_1} \sqrt{R_L} g_m \sqrt{\frac{1}{g_m}}}{C_L R_L + L_1 g_m}$
 wo: $\frac{\sqrt{\frac{1}{g_m}}}{\sqrt{C_L} \sqrt{L_1} \sqrt{R_L}}$
 bandwidth: $\frac{C_L R_L + L_1 g_m}{C_L L_1 R_L g_m}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{L_1 R_L g_m}{C_L R_L + L_1 g_m}$
 Qz: None
 Wz: None

3.3 BP-3 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \infty, R_L \right)$

$$H(s) = \frac{L_1 R_L g_m s}{C_1 L_1 s^2 + L_1 g_m s + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1}}{\sqrt{L_1} g_m}$
 wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
 bandwidth: $\frac{g_m}{C_1}$
 K-LP: 0
 K-HP: 0
 K-BP: R_L
 Qz: None
 Wz: None

3.4 BP-4 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \infty, \infty, R_L \right)$

Parameters:

Q: $\frac{\sqrt{C_1} R_1}{\sqrt{L_1 R_1 g_m} + \sqrt{L_1}}$
 wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
 bandwidth: $\frac{\sqrt{L_1 R_1 g_m} + \sqrt{L_1}}{C_1 \sqrt{L_1 R_1}}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_1 R_L g_m}{R_1 g_m + 1}$
 Qz: None
 Wz: None

4 BS

4.1 BS-1 $Z(s) = \left(R_1, \infty, \infty, \infty, \infty, \frac{R_L (C_L L_L s^2 + 1)}{C_L L_L s^2 + C_L R_L s + 1} \right)$

Parameters:

Q: $\frac{\sqrt{L_L}}{\sqrt{C_L} R_L}$
 wo: $\frac{1}{\sqrt{C_L} \sqrt{L_L}}$
 bandwidth: $\frac{R_L}{L_L}$
 K-LP: $\frac{R_1 R_L g_m}{R_1 g_m + 1}$
 K-HP: $\frac{R_1 R_L g_m}{R_1 g_m + 1}$
 K-BP: 0
 Qz: None
 Wz: $\frac{1}{\sqrt{C_L} \sqrt{L_L}}$

4.2 BS-2 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \infty, \infty, R_L \right)$

Parameters:

Q: $\frac{\sqrt{L_1 g_m}}{\sqrt{C_1}}$
 wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
 bandwidth: $\frac{1}{L_1 g_m}$
 K-LP: R_L
 K-HP: R_L
 K-BP: 0
 Qz: None
 Wz: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$

$$H(s) = \frac{L_1 R_1 R_L g_m s}{C_1 L_1 R_1 s^2 + R_1 + s (L_1 R_1 g_m + L_1)}$$

$$H(s) = \frac{C_L L_L R_1 R_L g_m s^2 + R_1 R_L g_m}{R_1 g_m + s^2 (C_L L_L R_1 g_m + C_L L_L) + s (C_L R_1 R_L g_m + C_L R_L) + 1}$$

$$H(s) = \frac{C_1 L_1 R_L g_m s^2 + R_L g_m}{C_1 L_1 g_m s^2 + C_1 s + g_m}$$

4.3 BS-3 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \infty, \infty, R_L \right)$

Parameters:

Q: $\frac{\sqrt{L_1} R_1 g_m + \sqrt{L_1}}{\sqrt{C_1} R_1}$
 wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
 bandwidth: $\frac{R_1}{\sqrt{L_1} (\sqrt{L_1} R_1 g_m + \sqrt{L_1})}$
 K-LP: $\frac{R_1 R_L g_m}{R_1 g_m + 1}$
 K-HP: $\frac{R_1 R_L g_m}{R_1 g_m + 1}$
 K-BP: 0
 Qz: None
 Wz: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$

5 GE

5.1 GE-1 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \infty, R_L \right)$

Parameters:

Q: $\frac{\sqrt{L_1} g_m}{\sqrt{C_1} R_1 g_m + \sqrt{C_1}}$
 wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
 bandwidth: $\frac{\sqrt{C_1} R_1 g_m + \sqrt{C_1}}{\sqrt{C_1} L_1 g_m}$
 K-LP: R_L
 K-HP: R_L
 K-BP: $\frac{R_1 R_L g_m}{R_1 g_m + 1}$
 Qz: $\frac{\sqrt{L_1}}{\sqrt{C_1} R_1}$
 Wz: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$

5.2 GE-2 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \infty, R_L \right)$

Parameters:

Q: $\frac{\sqrt{C_1} R_1 g_m + \sqrt{C_1}}{\sqrt{L_1} g_m}$
 wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
 bandwidth: $\frac{g_m}{\sqrt{C_1} (\sqrt{C_1} R_1 g_m + \sqrt{C_1})}$
 K-LP: $\frac{R_1 R_L g_m}{R_1 g_m + 1}$
 K-HP: $\frac{R_1 R_L g_m}{R_1 g_m + 1}$
 K-BP: R_L
 Qz: $\frac{\sqrt{C_1} R_1}{\sqrt{L_1}}$
 Wz: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$

6 HP

$$H(s) = \frac{C_1 L_1 R_1 R_L g_m s^2 + R_1 R_L g_m}{C_1 R_1 s + R_1 g_m + s^2 (C_1 L_1 R_1 g_m + C_1 L_1) + 1}$$

$$H(s) = \frac{C_1 L_1 R_L g_m s^2 + C_1 R_1 R_L g_m s + R_L g_m}{C_1 L_1 g_m s^2 + g_m + s (C_1 R_1 g_m + C_1)}$$

$$H(s) = \frac{C_1 L_1 R_1 R_L g_m s^2 + L_1 R_L g_m s + R_1 R_L g_m}{L_1 g_m s + R_1 g_m + s^2 (C_1 L_1 R_1 g_m + C_1 L_1) + 1}$$

7 LP

7.1 LP-1 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_L g_m}{C_1 C_L R_L s^2 + g_m + s(C_1 + C_L R_L g_m)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_L} \sqrt{R_L} \sqrt{g_m}}{C_1 + C_L R_L g_m}$
 wo: $\frac{\sqrt{g_m}}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_L}}$
 bandwidth: $\frac{C_1 + C_L R_L g_m}{C_1 C_L R_L}$
 K-LP: R_L
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.2 LP-2 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_1 R_L g_m}{C_1 C_L R_1 R_L s^2 + R_1 g_m + s(C_1 R_1 + C_L R_1 R_L g_m + C_L R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_L} \sqrt{R_1 g_m + 1}}{C_1 R_1 + C_L R_1 R_L g_m + C_L R_L}$
 wo: $\frac{\sqrt{R_1 g_m + 1}}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_L}}$
 bandwidth: $\frac{C_1 R_1 + C_L R_1 R_L g_m + C_L R_L}{C_1 C_L R_1 R_L}$
 K-LP: $\frac{R_1 R_L g_m}{R_1 g_m + 1}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.3 LP-3 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{L_1 g_m}{C_1 C_L L_1 s^2 + C_L L_1 g_m s + C_L}$$

Parameters:

Q: $\frac{\sqrt{C_1}}{\sqrt{L_1} g_m}$
 wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
 bandwidth: $\frac{g_m}{C_1}$
 K-LP: $\frac{L_1 g_m}{C_L}$
 K-HP: 0
 K-BP: 0
 Qz: None
 Wz: None

7.4 LP-4 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \infty, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{L_1 R_1 g_m}{C_1 C_L L_1 R_1 s^2 + C_L R_1 + s(C_L L_1 R_1 g_m + C_L L_1)}$$

Parameters:

Q: $\frac{\sqrt{C_1} R_1}{\sqrt{L_1} R_1 g_m + \sqrt{L_1}}$

wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{\sqrt{L_1}R_1g_m+\sqrt{L_1}}{C_1\sqrt{L_1}R_1}$
K-LP: $\frac{L_1g_m}{C_L}$
K-HP: 0
K-BP: 0
Qz: None
Wz: None

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(R_1 + \frac{1}{C_1s}, \infty, \infty, \infty, \infty, \frac{R_L}{C_LR_Ls+1} \right)$

$$H(s) = \frac{C_1R_1R_Lg_ms + R_Lg_m}{g_m + s^2(C_1C_LR_1R_Lg_m + C_1C_LR_L) + s(C_1R_1g_m + C_1 + C_LR_Lg_m)}$$

Parameters:

Q: $\frac{\sqrt{C_1}\sqrt{C_L}R_1\sqrt{R_Lg_m}\sqrt{\frac{g_m}{R_1g_m+1}}+\sqrt{C_1}\sqrt{C_L}\sqrt{R_L}\sqrt{\frac{g_m}{R_1g_m+1}}}{C_1R_1g_m+C_1+C_LR_Lg_m}$
wo: $\sqrt{\frac{g_m}{C_1C_LR_1R_Lg_m+C_1C_LR_L}}$
bandwidth: $\frac{\sqrt{\frac{g_m}{C_1C_LR_1R_Lg_m+C_1C_LR_L}}(C_1R_1g_m+C_1+C_LR_Lg_m)}{\sqrt{C_1}\sqrt{C_L}R_1\sqrt{R_Lg_m}\sqrt{\frac{g_m}{R_1g_m+1}}+\sqrt{C_1}\sqrt{C_L}\sqrt{R_L}\sqrt{\frac{g_m}{R_1g_m+1}}}$
K-LP: R_L
K-HP: 0
K-BP: $\frac{C_1R_1R_Lg_m}{C_1R_1g_m+C_1+C_LR_Lg_m}$
Qz: None
Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(\frac{L_1s}{C_1L_1s^2+1}, \infty, \infty, \infty, \infty, R_L + \frac{1}{C_Ls} \right)$

$$H(s) = \frac{C_LL_1R_Lg_ms + L_1g_m}{C_1C_LL_1s^2 + C_LL_1g_ms + C_L}$$

Parameters:

Q: $\frac{\sqrt{C_1}}{\sqrt{L_1}g_m}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{g_m}{C_1}$
K-LP: $\frac{L_1g_m}{C_L}$
K-HP: 0
K-BP: R_L
Qz: None
Wz: None

8.3 X-INVALID-NUMER-3 $Z(s) = \left(\frac{L_1R_1s}{C_1L_1R_1s^2+L_1s+R_1}, \infty, \infty, \infty, \infty, R_L + \frac{1}{C_Ls} \right)$

$$H(s) = \frac{C_LL_1R_1R_Lg_ms + L_1R_1g_m}{C_1C_LL_1R_1s^2 + C_LR_1 + s(C_LL_1R_1g_m + C_LL_1)}$$

Parameters:

Q: $\frac{\sqrt{C_1}R_1}{\sqrt{L_1}R_1g_m+\sqrt{L_1}}$
wo: $\frac{1}{\sqrt{C_1}\sqrt{L_1}}$
bandwidth: $\frac{\sqrt{L_1}R_1g_m+\sqrt{L_1}}{C_1\sqrt{L_1}R_1}$
K-LP: $\frac{L_1g_m}{C_L}$
K-HP: 0
K-BP: $\frac{R_1R_Lg_m}{R_1g_m+1}$

Qz: None
Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (R_1, \infty, \infty, \infty, \infty, R_L)$

$$H(s) = \frac{R_1 R_L g_m}{R_1 g_m + 1}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(R_1, \infty, \infty, \infty, \infty, \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 g_m}{s(C_L R_1 g_m + C_L)}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(R_1, \infty, \infty, \infty, \infty, \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{R_1 R_L g_m}{R_1 g_m + s(C_L R_1 R_L g_m + C_L R_L) + 1}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(R_1, \infty, \infty, \infty, \infty, R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_L R_1 R_L g_m s + R_1 g_m}{s(C_L R_1 g_m + C_L)}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(R_1, \infty, \infty, \infty, \infty, L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_L L_L R_1 g_m s^2 + R_1 g_m}{s(C_L R_1 g_m + C_L)}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(R_1, \infty, \infty, \infty, \infty, \frac{L_L s}{C_L L_L s^2 + 1}\right)$

$$H(s) = \frac{L_L R_1 g_m s}{R_1 g_m + s^2(C_L L_L R_1 g_m + C_L L_L) + 1}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(R_1, \infty, \infty, \infty, \infty, L_L s + R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{C_L L_L R_1 g_m s^2 + C_L R_1 R_L g_m s + R_1 g_m}{s(C_L R_1 g_m + C_L)}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(R_1, \infty, \infty, \infty, \infty, \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1}\right)$

$$H(s) = \frac{C_L L_L R_1 R_L g_m s^2 + L_L R_1 g_m s + R_1 R_L g_m}{R_1 g_m + s^2(C_L L_L R_1 g_m + C_L L_L) + 1}$$

$$\mathbf{9.9 \quad X-INVALID-ORDER-9} \quad Z(s) = (L_1s, \infty, \infty, \infty, \infty, R_L)$$

$$H(s) = \frac{L_1R_Lg_ms}{L_1g_ms + 1}$$

$$\mathbf{9.10 \quad X-INVALID-ORDER-10} \quad Z(s) = \left(L_1s, \infty, \infty, \infty, \infty, \frac{1}{C_Ls}\right)$$

$$H(s) = \frac{L_1g_m}{C_LL_1g_ms + C_L}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(L_1s, \infty, \infty, \infty, \infty, R_L + \frac{1}{C_Ls}\right)$$

$$H(s) = \frac{C_LL_1R_Lg_ms + L_1g_m}{C_LL_1g_ms + C_L}$$

$$\mathbf{9.12 \quad X-INVALID-ORDER-12} \quad Z(s) = \left(L_1s, \infty, \infty, \infty, \infty, L_Ls + \frac{1}{C_Ls}\right)$$

$$H(s) = \frac{C_LL_1L_Lg_ms^2 + L_1g_m}{C_LL_1g_ms + C_L}$$

$$\mathbf{9.13 \quad X-INVALID-ORDER-13} \quad Z(s) = \left(L_1s, \infty, \infty, \infty, \infty, \frac{L_Ls}{C_LL_Ls^2+1}\right)$$

$$H(s) = \frac{L_1L_Lg_ms^2}{C_LL_1L_Lg_ms^3 + C_LL_Ls^2 + L_1g_ms + 1}$$

$$\mathbf{9.14 \quad X-INVALID-ORDER-14} \quad Z(s) = \left(L_1s, \infty, \infty, \infty, \infty, L_Ls + R_L + \frac{1}{C_Ls}\right)$$

$$H(s) = \frac{C_LL_1L_Lg_ms^2 + C_LL_1R_Lg_ms + L_1g_m}{C_LL_1g_ms + C_L}$$

$$\mathbf{9.15 \quad X-INVALID-ORDER-15} \quad Z(s) = \left(L_1s, \infty, \infty, \infty, \infty, \frac{L_LR_Ls}{C_LL_LR_Ls^2+L_Ls+R_L}\right)$$

$$H(s) = \frac{L_1L_LR_Lg_ms^2}{C_LL_1L_LR_Lg_ms^3 + R_L + s^2(C_LL_LR_L + L_1L_Lg_m) + s(L_1R_Lg_m + L_L)}$$

$$\mathbf{9.16 \quad X-INVALID-ORDER-16} \quad Z(s) = \left(L_1s, \infty, \infty, \infty, \infty, \frac{C_LL_LR_Ls^2+L_Ls+R_L}{C_LL_Ls^2+1}\right)$$

$$H(s) = \frac{C_LL_1L_LR_Lg_ms^3 + L_1L_Lg_ms^2 + L_1R_Lg_ms}{C_LL_1L_Lg_ms^3 + C_LL_Ls^2 + L_1g_ms + 1}$$

$$\mathbf{9.17 \quad X-INVALID-ORDER-17} \quad Z(s) = \left(L_1s, \infty, \infty, \infty, \infty, \frac{R_L(C_LL_Ls^2+1)}{C_LL_Ls^2+C_LR_Ls+1}\right)$$

$$H(s) = \frac{C_LL_1L_LR_Lg_ms^3 + L_1R_Lg_ms}{C_LL_1L_Lg_ms^3 + s^2(C_LL_1R_Lg_m + C_LL_L) + s(C_LR_L + L_1g_m) + 1}$$

$$\mathbf{9.18 \quad X-INVALID-ORDER-18} \quad Z(s) = \left(\frac{1}{C_1s}, \infty, \infty, \infty, \infty, R_L\right)$$

$$H(s) = \frac{R_Lg_m}{C_1s + g_m}$$

$$\mathbf{9.19 \quad X-INVALID-ORDER-19} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \infty, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m}{C_1 C_L s^2 + C_L g_m s}$$

$$\mathbf{9.20 \quad X-INVALID-ORDER-20} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \infty, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_L R_L g_m s + g_m}{C_1 C_L s^2 + C_L g_m s}$$

$$\mathbf{9.21 \quad X-INVALID-ORDER-21} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \infty, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_L L_L g_m s^2 + g_m}{C_1 C_L s^2 + C_L g_m s}$$

$$\mathbf{9.22 \quad X-INVALID-ORDER-22} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{L_L g_m s}{C_1 C_L L_L s^3 + C_1 s + C_L L_L g_m s^2 + g_m}$$

$$\mathbf{9.23 \quad X-INVALID-ORDER-23} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \infty, L_L s + R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_L L_L g_m s^2 + C_L R_L g_m s + g_m}{C_1 C_L s^2 + C_L g_m s}$$

$$\mathbf{9.24 \quad X-INVALID-ORDER-24} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \infty, \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L} \right)$$

$$H(s) = \frac{L_L R_L g_m s}{C_1 C_L L_L R_L s^3 + R_L g_m + s^2 (C_1 L_L + C_L L_L R_L g_m) + s (C_1 R_L + L_L g_m)}$$

$$\mathbf{9.25 \quad X-INVALID-ORDER-25} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \infty, \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{C_L L_L R_L g_m s^2 + L_L g_m s + R_L g_m}{C_1 C_L L_L s^3 + C_1 s + C_L L_L g_m s^2 + g_m}$$

$$\mathbf{9.26 \quad X-INVALID-ORDER-26} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \infty, \frac{R_L (C_L L_L s^2 + 1)}{C_L L_L s^2 + C_L R_L s + 1} \right)$$

$$H(s) = \frac{C_L L_L R_L g_m s^2 + R_L g_m}{C_1 C_L L_L s^3 + g_m + s^2 (C_1 C_L R_L + C_L L_L g_m) + s (C_1 + C_L R_L g_m)}$$

$$\mathbf{9.27 \quad X-INVALID-ORDER-27} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \infty, R_L \right)$$

$$H(s) = \frac{R_1 R_L g_m}{C_1 R_1 s + R_1 g_m + 1}$$

$$\mathbf{9.28 \quad X-INVALID-ORDER-28} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \infty, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m}{C_1 C_L R_1 s^2 + s (C_L R_1 g_m + C_L)}$$

$$\mathbf{9.29 \quad X-INVALID-ORDER-29} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \infty, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_L R_1 R_L g_m s + R_1 g_m}{C_1 C_L R_1 s^2 + s (C_L R_1 g_m + C_L)}$$

$$\mathbf{9.30 \quad X-INVALID-ORDER-30} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \infty, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_L L_L R_1 g_m s^2 + R_1 g_m}{C_1 C_L R_1 s^2 + s (C_L R_1 g_m + C_L)}$$

$$\mathbf{9.31 \quad X-INVALID-ORDER-31} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{L_L R_1 g_m s}{C_1 C_L L_L R_1 s^3 + C_1 R_1 s + R_1 g_m + s^2 (C_L L_L R_1 g_m + C_L L_L) + 1}$$

$$\mathbf{9.32 \quad X-INVALID-ORDER-32} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \infty, L_L s + R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_L L_L R_1 g_m s^2 + C_L R_1 R_L g_m s + R_1 g_m}{C_1 C_L R_1 s^2 + s (C_L R_1 g_m + C_L)}$$

$$\mathbf{9.33 \quad X-INVALID-ORDER-33} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \infty, \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L} \right)$$

$$H(s) = \frac{L_L R_1 R_L g_m s}{C_1 C_L L_L R_1 R_L s^3 + R_1 R_L g_m + R_L + s^2 (C_1 L_L R_1 + C_L L_L R_1 R_L g_m + C_L L_L R_L) + s (C_1 R_1 R_L + L_L R_1 g_m + L_L)}$$

$$\mathbf{9.34 \quad X-INVALID-ORDER-34} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \infty, \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{C_L L_L R_1 R_L g_m s^2 + L_L R_1 g_m s + R_1 R_L g_m}{C_1 C_L L_L R_1 s^3 + C_1 R_1 s + R_1 g_m + s^2 (C_L L_L R_1 g_m + C_L L_L) + 1}$$

$$\mathbf{9.35 \quad X-INVALID-ORDER-35} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \infty, \frac{R_L (C_L L_L s^2 + 1)}{C_L L_L s^2 + C_L R_L s + 1} \right)$$

$$H(s) = \frac{C_L L_L R_1 R_L g_m s^2 + R_1 R_L g_m}{C_1 C_L L_L R_1 s^3 + R_1 g_m + s^2 (C_1 C_L R_1 R_L + C_L L_L R_1 g_m + C_L L_L) + s (C_1 R_1 + C_L R_1 R_L g_m + C_L R_L) + 1}$$

$$\mathbf{9.36 \quad X-INVALID-ORDER-36} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \infty, R_L \right)$$

$$H(s) = \frac{C_1 R_1 R_L g_m s + R_L g_m}{g_m + s (C_1 R_1 g_m + C_1)}$$

$$\mathbf{9.37 \quad X-INVALID-ORDER-37} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \infty, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_1 R_1 g_m s + g_m}{C_L g_m s + s^2 (C_1 C_L R_1 g_m + C_1 C_L)}$$

$$\mathbf{9.38 \quad X-INVALID-ORDER-38} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \infty, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_1 C_L R_1 R_L g_m s^2 + g_m + s (C_1 R_1 g_m + C_L R_L g_m)}{C_L g_m s + s^2 (C_1 C_L R_1 g_m + C_1 C_L)}$$

$$\mathbf{9.39 \quad X-INVALID-ORDER-39} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \infty, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_1 C_L L_L R_1 g_m s^3 + C_1 R_1 g_m s + C_L L_L g_m s^2 + g_m}{C_L g_m s + s^2 (C_1 C_L R_1 g_m + C_1 C_L)}$$

$$\mathbf{9.40 \quad X-INVALID-ORDER-40} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{C_1 L_L R_1 g_m s^2 + L_L g_m s}{C_L L_L g_m s^2 + g_m + s^3 (C_1 C_L L_L R_1 g_m + C_1 C_L L_L) + s (C_1 R_1 g_m + C_1)}$$

$$\mathbf{9.41 \quad X-INVALID-ORDER-41} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \infty, \quad L_L s + R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_1 C_L L_L R_1 g_m s^3 + g_m + s^2 (C_1 C_L R_1 R_L g_m + C_L L_L g_m) + s (C_1 R_1 g_m + C_L R_L g_m)}{C_L g_m s + s^2 (C_1 C_L R_1 g_m + C_1 C_L)}$$

$$\mathbf{9.42 \quad X-INVALID-ORDER-42} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L} \right)$$

$$H(s) = \frac{C_1 L_L R_1 R_L g_m s^2 + L_L R_L g_m s}{R_L g_m + s^3 (C_1 C_L L_L R_1 R_L g_m + C_1 C_L L_L R_L) + s^2 (C_1 L_L R_1 g_m + C_1 L_L + C_L L_L R_L g_m) + s (C_1 R_1 R_L g_m + C_1 R_L + L_L g_m)}$$

$$\mathbf{9.43 \quad X-INVALID-ORDER-43} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \infty, \quad \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{C_1 C_L L_L R_1 R_L g_m s^3 + R_L g_m + s^2 (C_1 L_L R_1 g_m + C_L L_L R_L g_m) + s (C_1 R_1 R_L g_m + L_L g_m)}{C_L L_L g_m s^2 + g_m + s^3 (C_1 C_L L_L R_1 g_m + C_1 C_L L_L) + s (C_1 R_1 g_m + C_1)}$$

$$\mathbf{9.44 \quad X-INVALID-ORDER-44} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_L (C_L L_L s^2 + 1)}{C_L L_L s^2 + C_L R_L s + 1} \right)$$

$$H(s) = \frac{C_1 C_L L_L R_1 R_L g_m s^3 + C_1 R_1 R_L g_m s + C_L L_L R_L g_m s^2 + R_L g_m}{g_m + s^3 (C_1 C_L L_L R_1 g_m + C_1 C_L L_L) + s^2 (C_1 C_L R_1 R_L g_m + C_1 C_L R_L + C_L L_L g_m) + s (C_1 R_1 g_m + C_1 + C_L R_L g_m)}$$

$$\mathbf{9.45 \quad X-INVALID-ORDER-45} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_1 L_1 g_m s^2 + g_m}{C_1 C_L L_1 g_m s^3 + C_1 C_L s^2 + C_L g_m s}$$

$$\mathbf{9.46 \quad X-INVALID-ORDER-46} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{C_1 L_1 R_L g_m s^2 + R_L g_m}{C_1 C_L L_1 R_L g_m s^3 + g_m + s^2 (C_1 C_L R_L + C_1 L_1 g_m) + s (C_1 + C_L R_L g_m)}$$

$$\mathbf{9.47 \quad X-INVALID-ORDER-47} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \infty, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_1 C_L L_1 R_L g_m s^3 + C_1 L_1 g_m s^2 + C_L R_L g_m s + g_m}{C_1 C_L L_1 g_m s^3 + C_1 C_L s^2 + C_L g_m s}$$

$$\mathbf{9.48 \quad X-INVALID-ORDER-48} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \infty, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_1 C_L L_1 L_L g_m s^4 + g_m + s^2 (C_1 L_1 g_m + C_L L_L g_m)}{C_1 C_L L_1 g_m s^3 + C_1 C_L s^2 + C_L g_m s}$$

$$\mathbf{9.49 \quad X-INVALID-ORDER-49} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{C_1 L_1 L_L g_m s^3 + L_L g_m s}{C_1 C_L L_1 L_L g_m s^4 + C_1 C_L L_L s^3 + C_1 s + g_m + s^2 (C_1 L_1 g_m + C_L L_L g_m)}$$

$$\mathbf{9.50 \quad X-INVALID-ORDER-50} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \infty, \quad L_L s + R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_1 C_L L_1 L_L g_m s^4 + C_1 C_L L_1 R_L g_m s^3 + C_L R_L g_m s + g_m + s^2 (C_1 L_1 g_m + C_L L_L g_m)}{C_1 C_L L_1 g_m s^3 + C_1 C_L s^2 + C_L g_m s}$$

$$\mathbf{9.51 \quad X-INVALID-ORDER-51} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L} \right)$$

$$H(s) = \frac{C_1 L_1 L_L R_L g_m s^3 + L_L R_L g_m s}{C_1 C_L L_1 L_L R_L g_m s^4 + R_L g_m + s^3 (C_1 C_L L_L R_L + C_1 L_1 L_L g_m) + s^2 (C_1 L_1 R_L g_m + C_1 L_L + C_L L_L R_L g_m) + s (C_1 R_L + L_L g_m)}$$

$$\mathbf{9.52 \quad X-INVALID-ORDER-52} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \infty, \quad \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{C_1 C_L L_1 L_L R_L g_m s^4 + C_1 L_1 L_L g_m s^3 + L_L g_m s + R_L g_m + s^2 (C_1 L_1 R_L g_m + C_L L_L R_L g_m)}{C_1 C_L L_1 L_L g_m s^4 + C_1 C_L L_L s^3 + C_1 s + g_m + s^2 (C_1 L_1 g_m + C_L L_L g_m)}$$

$$\mathbf{9.53 \quad X-INVALID-ORDER-53} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_L (C_L L_L s^2 + 1)}{C_L L_L s^2 + C_L R_L s + 1} \right)$$

$$H(s) = \frac{C_1 C_L L_1 L_L R_L g_m s^4 + R_L g_m + s^2 (C_1 L_1 R_L g_m + C_L L_L R_L g_m)}{C_1 C_L L_1 L_L g_m s^4 + g_m + s^3 (C_1 C_L L_1 R_L g_m + C_1 C_L L_L) + s^2 (C_1 C_L R_L + C_1 L_1 g_m + C_L L_L g_m) + s (C_1 + C_L R_L g_m)}$$

$$\mathbf{9.54 \quad X-INVALID-ORDER-54} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{L_1 R_L g_m s}{C_1 C_L L_1 R_L s^3 + s^2 (C_1 L_1 + C_L L_1 R_L g_m) + s (C_L R_L + L_1 g_m) + 1}$$

$$\mathbf{9.55 \quad X-INVALID-ORDER-55} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{L_1 L_L g_m s^2}{C_1 C_L L_1 L_L s^4 + C_L L_1 L_L g_m s^3 + L_1 g_m s + s^2 (C_1 L_1 + C_L L_L) + 1}$$

$$\mathbf{9.56 \quad X-INVALID-ORDER-56} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L} \right)$$

$$H(s) = \frac{L_1 L_L R_L g_m s^2}{C_1 C_L L_1 L_L R_L s^4 + R_L + s^3 (C_1 L_1 L_L + C_L L_1 L_L R_L g_m) + s^2 (C_1 L_1 R_L + C_L L_L R_L + L_1 L_L g_m) + s (L_1 R_L g_m + L_L)}$$

$$\mathbf{9.57 \quad X-INVALID-ORDER-57} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \infty, \quad \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{C_L L_1 L_L R_L g_m s^3 + L_1 L_L g_m s^2 + L_1 R_L g_m s}{C_1 C_L L_1 L_L s^4 + C_L L_1 L_L g_m s^3 + L_1 g_m s + s^2 (C_1 L_1 + C_L L_L) + 1}$$

$$\mathbf{9.58 \quad X-INVALID-ORDER-58} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \infty, \frac{R_L (C_L L_L s^2 + 1)}{C_L L_L s^2 + C_L R_L s + 1} \right)$$

$$H(s) = \frac{C_L L_1 L_L R_L g_m s^3 + L_1 R_L g_m s}{C_1 C_L L_1 L_L s^4 + s^3 (C_1 C_L L_1 R_L + C_L L_1 L_L g_m) + s^2 (C_1 L_1 + C_L L_1 R_L g_m + C_L L_L) + s (C_L R_L + L_1 g_m) + 1}$$

$$\mathbf{9.59 \quad X-INVALID-ORDER-59} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \infty, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_1 L_1 g_m s^2 + C_1 R_1 g_m s + g_m}{C_1 C_L L_1 g_m s^3 + C_L g_m s + s^2 (C_1 C_L R_1 g_m + C_1 C_L)}$$

$$\mathbf{9.60 \quad X-INVALID-ORDER-60} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{C_1 L_1 R_L g_m s^2 + C_1 R_1 R_L g_m s + R_L g_m}{C_1 C_L L_1 R_L g_m s^3 + g_m + s^2 (C_1 C_L R_1 R_L g_m + C_1 C_L R_L + C_1 L_1 g_m) + s (C_1 R_1 g_m + C_1 + C_L R_L g_m)}$$

$$\mathbf{9.61 \quad X-INVALID-ORDER-61} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \infty, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_1 C_L L_1 R_L g_m s^3 + g_m + s^2 (C_1 C_L R_1 R_L g_m + C_1 L_1 g_m) + s (C_1 R_1 g_m + C_L R_L g_m)}{C_1 C_L L_1 g_m s^3 + C_L g_m s + s^2 (C_1 C_L R_1 g_m + C_1 C_L)}$$

$$\mathbf{9.62 \quad X-INVALID-ORDER-62} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \infty, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_1 C_L L_1 L_L g_m s^4 + C_1 C_L L_L R_1 g_m s^3 + C_1 R_1 g_m s + g_m + s^2 (C_1 L_1 g_m + C_L L_L g_m)}{C_1 C_L L_1 g_m s^3 + C_L g_m s + s^2 (C_1 C_L R_1 g_m + C_1 C_L)}$$

$$\mathbf{9.63 \quad X-INVALID-ORDER-63} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{C_1 L_1 L_L g_m s^3 + C_1 L_L R_1 g_m s^2 + L_L g_m s}{C_1 C_L L_1 L_L g_m s^4 + g_m + s^3 (C_1 C_L L_L R_1 g_m + C_1 C_L L_L) + s^2 (C_1 L_1 g_m + C_L L_L g_m) + s (C_1 R_1 g_m + C_1)}$$

$$\mathbf{9.64 \quad X-INVALID-ORDER-64} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \infty, L_L s + R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_1 C_L L_1 L_L g_m s^4 + g_m + s^3 (C_1 C_L L_1 R_L g_m + C_1 C_L L_L R_1 g_m) + s^2 (C_1 C_L R_1 R_L g_m + C_1 L_1 g_m + C_L L_L g_m) + s (C_1 R_1 g_m + C_L R_L g_m)}{C_1 C_L L_1 g_m s^3 + C_L g_m s + s^2 (C_1 C_L R_1 g_m + C_1 C_L)}$$

$$\mathbf{9.65 \quad X-INVALID-ORDER-65} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \infty, \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L} \right)$$

$$H(s) = \frac{C_1 L_1 L_L R_L g_m s^3 + C_1 L_L R_1 R_L g_m s^2 + L_L R_L g_m s}{C_1 C_L L_1 L_L R_L g_m s^4 + R_L g_m + s^3 (C_1 C_L L_L R_1 R_L g_m + C_1 C_L L_L R_L + C_1 L_1 L_L g_m) + s^2 (C_1 L_1 R_L g_m + C_1 L_L R_1 g_m + C_1 L_L + C_L L_L R_L g_m) + s (C_1 R_1 R_L g_m + C_1 R_L + L_L g_m)}$$

$$\mathbf{9.66 \quad X-INVALID-ORDER-66} \quad Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \infty, \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{C_1 C_L L_1 L_L R_L g_m s^4 + R_L g_m + s^3 (C_1 C_L L_L R_1 R_L g_m + C_1 L_1 L_L g_m) + s^2 (C_1 L_1 R_L g_m + C_1 L_L R_1 g_m + C_L L_L R_L g_m) + s (C_1 R_1 R_L g_m + L_L g_m)}{C_1 C_L L_1 L_L g_m s^4 + g_m + s^3 (C_1 C_L L_L R_1 g_m + C_1 C_L L_L) + s^2 (C_1 L_1 g_m + C_L L_L g_m) + s (C_1 R_1 g_m + C_1)}$$

9.67 X-INVALID-ORDER-67 $Z(s) = \left(L_1 s + R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \infty, \frac{R_L(C_L L_L s^2 + 1)}{C_L L_L s^2 + C_L R_L s + 1} \right)$

$$H(s) = \frac{C_1 C_L L_1 L_L R_L g_m s^4 + C_1 C_L L_L R_1 R_L g_m s^3 + C_1 R_1 R_L g_m s + R_L g_m + s^2 (C_1 L_1 R_L g_m + C_L L_L R_L g_m)}{C_1 C_L L_1 L_L g_m s^4 + g_m + s^3 (C_1 C_L L_1 R_L g_m + C_1 C_L L_L R_1 g_m + C_1 C_L L_L) + s^2 (C_1 C_L R_1 R_L g_m + C_1 C_L R_L + C_1 L_1 g_m + C_L L_L g_m) + s (C_1 R_1 g_m + C_1 + C_L R_L g_m)}$$

9.68 X-INVALID-ORDER-68 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{L_1 R_1 R_L g_m s}{C_1 C_L L_1 R_1 R_L s^3 + R_1 + s^2 (C_1 L_1 R_1 + C_L L_1 R_1 R_L g_m + C_L L_1 R_L) + s (C_L R_1 R_L + L_1 R_1 g_m + L_1)}$$

9.69 X-INVALID-ORDER-69 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \infty, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{L_1 L_L R_1 g_m s^2}{C_1 C_L L_1 L_L R_1 s^4 + R_1 + s^3 (C_L L_1 L_L R_1 g_m + C_L L_1 L_L) + s^2 (C_1 L_1 R_1 + C_L L_L R_1) + s (L_1 R_1 g_m + L_1)}$$

9.70 X-INVALID-ORDER-70 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \infty, \infty, \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L} \right)$

$$H(s) = \frac{L_1 L_L R_1 R_L g_m s^2}{C_1 C_L L_1 L_L R_1 R_L s^4 + R_1 R_L + s^3 (C_1 L_1 L_L R_1 + C_L L_1 L_L R_1 R_L g_m + C_L L_1 L_L R_L) + s^2 (C_1 L_1 R_1 R_L + C_L L_L R_1 R_L + L_1 L_L R_1 g_m + L_1 L_L) + s (L_1 R_1 R_L g_m + L_1 R_L + L_L R_1)}$$

9.71 X-INVALID-ORDER-71 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \infty, \infty, \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{C_L L_1 L_L R_1 R_L g_m s^3 + L_1 L_L R_1 g_m s^2 + L_1 R_1 R_L g_m s}{C_1 C_L L_1 L_L R_1 s^4 + R_1 + s^3 (C_L L_1 L_L R_1 g_m + C_L L_1 L_L) + s^2 (C_1 L_1 R_1 + C_L L_L R_1) + s (L_1 R_1 g_m + L_1)}$$

9.72 X-INVALID-ORDER-72 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \infty, \infty, \frac{R_L(C_L L_L s^2 + 1)}{C_L L_L s^2 + C_L R_L s + 1} \right)$

$$H(s) = \frac{C_L L_1 L_L R_1 R_L g_m s^3 + L_1 R_1 R_L g_m s}{C_1 C_L L_1 L_L R_1 s^4 + R_1 + s^3 (C_1 C_L L_1 R_1 R_L + C_L L_1 L_L R_1 g_m + C_L L_1 L_L) + s^2 (C_1 L_1 R_1 + C_L L_1 R_1 R_L g_m + C_L L_1 R_L + C_L L_L R_1) + s (C_L R_1 R_L + L_1 R_1 g_m + L_1)}$$

9.73 X-INVALID-ORDER-73 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \infty, \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_1 L_1 R_1 g_m s^2 + L_1 g_m s + R_1 g_m}{C_L L_1 g_m s^2 + s^3 (C_1 C_L L_1 R_1 g_m + C_1 C_L L_1) + s (C_L R_1 g_m + C_L)}$$

9.74 X-INVALID-ORDER-74 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{C_1 L_1 R_1 R_L g_m s^2 + L_1 R_L g_m s + R_1 R_L g_m}{R_1 g_m + s^3 (C_1 C_L L_1 R_1 R_L g_m + C_1 C_L L_1 R_L) + s^2 (C_1 L_1 R_1 g_m + C_1 L_1 + C_L L_1 R_L g_m) + s (C_L R_1 R_L g_m + C_L R_L + L_1 g_m) + 1}$$

9.75 X-INVALID-ORDER-75 $Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \infty, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_1 C_L L_1 R_1 R_L g_m s^3 + R_1 g_m + s^2 (C_1 L_1 R_1 g_m + C_L L_1 R_L g_m) + s (C_L R_1 R_L g_m + L_1 g_m)}{C_L L_1 g_m s^2 + s^3 (C_1 C_L L_1 R_1 g_m + C_1 C_L L_1) + s (C_L R_1 g_m + C_L)}$$

$$\mathbf{9.76 \quad X-INVALID-ORDER-76} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \infty, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_1 C_L L_1 L_L R_1 g_m s^4 + C_L L_1 L_L g_m s^3 + L_1 g_m s + R_1 g_m + s^2 (C_1 L_1 R_1 g_m + C_L L_L R_1 g_m)}{C_L L_1 g_m s^2 + s^3 (C_1 C_L L_1 R_1 g_m + C_1 C_L L_1) + s (C_L R_1 g_m + C_L)}$$

$$\mathbf{9.77 \quad X-INVALID-ORDER-77} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{C_1 L_1 L_L R_1 g_m s^3 + L_1 L_L g_m s^2 + L_L R_1 g_m s}{C_L L_1 L_L g_m s^3 + L_1 g_m s + R_1 g_m + s^4 (C_1 C_L L_1 L_L R_1 g_m + C_1 C_L L_1 L_L) + s^2 (C_1 L_1 R_1 g_m + C_1 L_1 + C_L L_L R_1 g_m + C_L L_L) + 1}$$

$$\mathbf{9.78 \quad X-INVALID-ORDER-78} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \infty, L_L s + R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_1 C_L L_1 L_L R_1 g_m s^4 + R_1 g_m + s^3 (C_1 C_L L_1 R_1 R_L g_m + C_L L_1 L_L g_m) + s^2 (C_1 L_1 R_1 g_m + C_L L_1 R_L g_m + C_L L_L R_1 g_m) + s (C_L R_1 R_L g_m + L_1 g_m)}{C_L L_1 g_m s^2 + s^3 (C_1 C_L L_1 R_1 g_m + C_1 C_L L_1) + s (C_L R_1 g_m + C_L)}$$

$$\mathbf{9.79 \quad X-INVALID-ORDER-79} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \infty, \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L} \right)$$

$$H(s) = \frac{C_1 L_1 L_L R_1 R_L g_m s^3 + L_1 L_L R_L g_m s^2 + L_L R_1 R_L g_m s}{R_1 R_L g_m + R_L + s^4 (C_1 C_L L_1 L_L R_1 R_L g_m + C_1 C_L L_1 L_L R_L) + s^3 (C_1 L_1 L_L R_1 g_m + C_1 L_1 L_L + C_L L_1 L_L R_L g_m) + s^2 (C_1 L_1 R_1 R_L g_m + C_1 L_1 R_L + C_L L_L R_1 R_L g_m + C_L L_L R_L + L_1 L_L g_m) + s (L_1 R_L g_m + L_L R_1 g_m + L_L)}$$

$$\mathbf{9.80 \quad X-INVALID-ORDER-80} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \infty, \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{C_1 C_L L_1 L_L R_1 R_L g_m s^4 + R_1 R_L g_m + s^3 (C_1 L_1 L_L R_1 g_m + C_L L_1 L_L R_L g_m) + s^2 (C_1 L_1 R_1 R_L g_m + C_L L_L R_1 R_L g_m + L_1 L_L g_m) + s (L_1 R_L g_m + L_L R_1 g_m)}{C_L L_1 L_L g_m s^3 + L_1 g_m s + R_1 g_m + s^4 (C_1 C_L L_1 L_L R_1 g_m + C_1 C_L L_1 L_L) + s^2 (C_1 L_1 R_1 g_m + C_1 L_1 + C_L L_L R_1 g_m + C_L L_L) + 1}$$

$$\mathbf{9.81 \quad X-INVALID-ORDER-81} \quad Z(s) = \left(\frac{C_1 L_1 R_1 s^2 + L_1 s + R_1}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \infty, \frac{R_L (C_L L_L s^2 + 1)}{C_L L_L s^2 + C_L R_L s + 1} \right)$$

$$H(s) = \frac{C_1 C_L L_1 L_L R_1 R_L g_m s^4 + C_L L_1 L_L R_L g_m s^3 + L_1 R_L g_m s + R_1 R_L g_m + s^2 (C_1 L_1 R_1 R_L g_m + C_L L_L R_1 R_L g_m)}{R_1 g_m + s^4 (C_1 C_L L_1 L_L R_1 g_m + C_1 C_L L_1 L_L) + s^3 (C_1 C_L L_1 R_1 R_L g_m + C_1 C_L L_1 R_L + C_L L_1 L_L g_m) + s^2 (C_1 L_1 R_1 g_m + C_1 L_1 + C_L L_1 R_L g_m + C_L L_L R_1 g_m + C_L L_L) + s (C_L R_1 R_L g_m + C_L R_L + L_1 g_m) + 1}$$

$$\mathbf{9.82 \quad X-INVALID-ORDER-82} \quad Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \infty, \infty, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_1 L_1 R_1 g_m s^2 + R_1 g_m}{C_1 C_L R_1 s^2 + s^3 (C_1 C_L L_1 R_1 g_m + C_1 C_L L_1) + s (C_L R_1 g_m + C_L)}$$

$$\mathbf{9.83 \quad X-INVALID-ORDER-83} \quad Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \infty, \infty, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{C_1 L_1 R_1 R_L g_m s^2 + R_1 R_L g_m}{R_1 g_m + s^3 (C_1 C_L L_1 R_1 R_L g_m + C_1 C_L L_1 R_L) + s^2 (C_1 C_L R_1 R_L + C_1 L_1 R_1 g_m + C_1 L_1) + s (C_1 R_1 + C_L R_1 R_L g_m + C_L R_L) + 1}$$

$$\mathbf{9.84 \quad X-INVALID-ORDER-84} \quad Z(s) = \left(\frac{R_1 (C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \infty, \infty, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{C_1 C_L L_1 R_1 R_L g_m s^3 + C_1 L_1 R_1 g_m s^2 + C_L R_1 R_L g_m s + R_1 g_m}{C_1 C_L R_1 s^2 + s^3 (C_1 C_L L_1 R_1 g_m + C_1 C_L L_1) + s (C_L R_1 g_m + C_L)}$$

9.85 X-INVALID-ORDER-85 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \infty, \infty, L_L s + \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_1 C_L L_1 L_L R_1 g_m s^4 + R_1 g_m + s^2 (C_1 L_1 R_1 g_m + C_L L_L R_1 g_m)}{C_1 C_L R_1 s^2 + s^3 (C_1 C_L L_1 R_1 g_m + C_1 C_L L_1) + s (C_L R_1 g_m + C_L)}$$

9.86 X-INVALID-ORDER-86 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \infty, \infty, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{C_1 L_1 L_L R_1 g_m s^3 + L_L R_1 g_m s}{C_1 C_L L_L R_1 s^3 + C_1 R_1 s + R_1 g_m + s^4 (C_1 C_L L_1 L_L R_1 g_m + C_1 C_L L_1 L_L) + s^2 (C_1 L_1 R_1 g_m + C_1 L_1 + C_L L_L R_1 g_m + C_L L_L) + 1}$$

9.87 X-INVALID-ORDER-87 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \infty, \infty, L_L s + R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_1 C_L L_1 L_L R_1 g_m s^4 + C_1 C_L L_1 R_1 R_L g_m s^3 + C_L R_1 R_L g_m s + R_1 g_m + s^2 (C_1 L_1 R_1 g_m + C_L L_L R_1 g_m)}{C_1 C_L R_1 s^2 + s^3 (C_1 C_L L_1 R_1 g_m + C_1 C_L L_1) + s (C_L R_1 g_m + C_L)}$$

9.88 X-INVALID-ORDER-88 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \infty, \infty, \frac{L_L R_L s}{C_L L_L R_L s^2 + L_L s + R_L} \right)$

$$H(s) = \frac{C_1 L_1 L_L R_1 R_L g_m s^3 + L_L R_1 R_L g_m s}{R_1 R_L g_m + R_L + s^4 (C_1 C_L L_1 L_L R_1 R_L g_m + C_1 C_L L_1 L_L R_L) + s^3 (C_1 C_L L_L R_1 R_L + C_1 L_1 L_L R_1 g_m + C_1 L_1 L_L) + s^2 (C_1 L_1 R_1 R_L g_m + C_1 L_1 R_L + C_1 L_L R_1 + C_L L_L R_1 R_L g_m + C_L L_L R_L) + s (C_1 R_1 R_L + L_L R_1 g_m + L_L)}$$

9.89 X-INVALID-ORDER-89 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \infty, \infty, \frac{C_L L_L R_L s^2 + L_L s + R_L}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{C_1 C_L L_1 L_L R_1 R_L g_m s^4 + C_1 L_1 L_L R_1 g_m s^3 + L_L R_1 g_m s + R_1 R_L g_m + s^2 (C_1 L_1 R_1 R_L g_m + C_L L_L R_1 R_L g_m)}{C_1 C_L L_L R_1 s^3 + C_1 R_1 s + R_1 g_m + s^4 (C_1 C_L L_1 L_L R_1 g_m + C_1 C_L L_1 L_L) + s^2 (C_1 L_1 R_1 g_m + C_1 L_1 + C_L L_L R_1 g_m + C_L L_L) + 1}$$

9.90 X-INVALID-ORDER-90 $Z(s) = \left(\frac{R_1(C_1 L_1 s^2 + 1)}{C_1 L_1 s^2 + C_1 R_1 s + 1}, \infty, \infty, \infty, \infty, \frac{R_L(C_L L_L s^2 + 1)}{C_L L_L s^2 + C_L R_L s + 1} \right)$

$$H(s) = \frac{C_1 C_L L_1 L_L R_1 R_L g_m s^4 + R_1 R_L g_m + s^2 (C_1 L_1 R_1 R_L g_m + C_L L_L R_1 R_L g_m)}{R_1 g_m + s^4 (C_1 C_L L_1 L_L R_1 g_m + C_1 C_L L_1 L_L) + s^3 (C_1 C_L L_1 R_1 R_L g_m + C_1 C_L L_1 R_L + C_1 C_L L_L R_1) + s^2 (C_1 C_L R_1 R_L + C_1 L_1 R_1 g_m + C_1 L_1 + C_L L_L R_1 g_m + C_L L_L) + s (C_1 R_1 + C_L R_1 R_L g_m + C_L R_L) + 1}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \infty, L_L s + \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_L L_1 L_L g_m s^2 + L_1 g_m}{C_1 C_L L_1 s^2 + C_L L_1 g_m s + C_L}$$

Parameters:

Q: $\frac{\sqrt{C_1}}{\sqrt{L_1 g_m}}$
 wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
 bandwidth: $\frac{g_m}{C_1}$
 K-LP: $\frac{L_1 g_m}{C_L}$
 K-HP: $\frac{L_L g_m}{C_1}$
 K-BP: 0
 Qz: None
 Wz: $\frac{1}{\sqrt{C_L} \sqrt{L_L}}$

10.2 X-INVALID-WZ-2 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \infty, L_L s + R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_L L_1 L_L g_m s^2 + C_L L_1 R_L g_m s + L_1 g_m}{C_1 C_L L_1 s^2 + C_L L_1 g_m s + C_L}$$

Parameters:

Q: $\frac{\sqrt{C_1}}{\sqrt{L_1 g_m}}$
 wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
 bandwidth: $\frac{g_m}{C_1}$
 K-LP: $\frac{L_1 g_m}{C_L}$
 K-HP: $\frac{L_L g_m}{C_1}$
 K-BP: R_L
 Qz: None
 Wz: $\frac{1}{\sqrt{C_L} \sqrt{L_L}}$

10.3 X-INVALID-WZ-3 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \infty, \infty, L_L s + \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_L L_1 L_L R_1 g_m s^2 + L_1 R_1 g_m}{C_1 C_L L_1 R_1 s^2 + C_L R_1 + s (C_L L_1 R_1 g_m + C_L L_1)}$$

Parameters:

Q: $\frac{\sqrt{C_1} R_1}{\sqrt{L_1 R_1 g_m + \sqrt{L_1}}}$
 wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
 bandwidth: $\frac{\sqrt{L_1 R_1 g_m + \sqrt{L_1}}}{C_1 \sqrt{L_1 R_1}}$
 K-LP: $\frac{L_1 g_m}{C_L}$
 K-HP: $\frac{L_L g_m}{C_1}$
 K-BP: 0
 Qz: None
 Wz: $\frac{1}{\sqrt{C_L} \sqrt{L_L}}$

10.4 X-INVALID-WZ-4 $Z(s) = \left(\frac{L_1 R_1 s}{C_1 L_1 R_1 s^2 + L_1 s + R_1}, \infty, \infty, \infty, \infty, L_L s + R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{C_L L_1 L_L R_1 g_m s^2 + C_L L_1 R_1 R_L g_m s + L_1 R_1 g_m}{C_1 C_L L_1 R_1 s^2 + C_L R_1 + s (C_L L_1 R_1 g_m + C_L L_1)}$$

Parameters:

Q: $\frac{\sqrt{C_1} R_1}{\sqrt{L_1 R_1 g_m + \sqrt{L_1}}}$
 wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
 bandwidth: $\frac{\sqrt{L_1 R_1 g_m + \sqrt{L_1}}}{C_1 \sqrt{L_1 R_1}}$
 K-LP: $\frac{L_1 g_m}{C_L}$
 K-HP: $\frac{L_L g_m}{C_1}$
 K-BP: $\frac{R_1 R_L g_m}{R_1 g_m + 1}$
 Qz: None
 Wz: $\frac{1}{\sqrt{C_L} \sqrt{L_L}}$

11 X-PolynomialError